IT5006 FUNDAMENTALS OF DATA ANALYTICS · GROUP 10 · FINAL PRESENTATION

Predicting delivery performance and satisfaction risk

Two prediction moments on the Olist Brazilian marketplace — from nine tables to a deployed application



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Live app **team10-it5006-olist.streamlit.app** Code **github.com/gengyuzccc-cell/Team10_IT5006_Ecommerce_Analytics_AY2627Sem1**

One number drove the whole project

A late delivery is the strongest single driver of a bad review in 99,091 Olist orders (2017–2018)

1–2 STAR REVIEW RATE

9.2% on time

54.0% delivered late

5.9× more likely, and monotone in how late the parcel is. Review score correlates −0.33 with lead time and ~0 with price or basket size: the delivery, not the purchase, drives the rating.

77% of reviews are 4–5 stars; the minority is what we predict. Lead time itself halves across the window, so every model also has to survive drift.

One decision chain: Problem 1's output is Problem 2's most important input.



PROBLEM 1 · AT CHECKOUT

How long will this order take, and will it miss its promise?

Regression on lead time + a late flag · Operations: re-quote, pre-warn, expedite

PROBLEM 2 · AT DELIVERY

Will this delivered order draw a 1–2 star review?

Classification · Customer experience: reach out before the review is written

Nine tables, one row per order — and two traps inside them

Chronological, customer-disjoint split; leakage defined per prediction moment

Join inflation

Items, payments and reviews are one-to-many. A naive merge turns 99,441 orders into 118,437 rows (+19%) and biases every average. We aggregate each child table first, then join, and assert the grain.

customer_id is not a person

99,441 customer_id values map to 96,096 people. Group on the wrong key and there are no repeat buyers; on the right one, 2,997 are — and 227 straddle the split, so we remove them.

Leakage, one rule per moment

At checkout nothing about fulfilment exists; at delivery the delivery outcome is history. assert_no_leakage(features, regime) runs before any fit.

ONE JOB PER PERIOD



Train Jan 2017 – Apr 2018 · 73,008 orders · fit, tune with expanding-window CV

Validation May – Jun 2018 · 13,156 · choose models, thresholds, calibration

Test Jul – Aug 2018 · 12,927 · scored once, replayed by the app

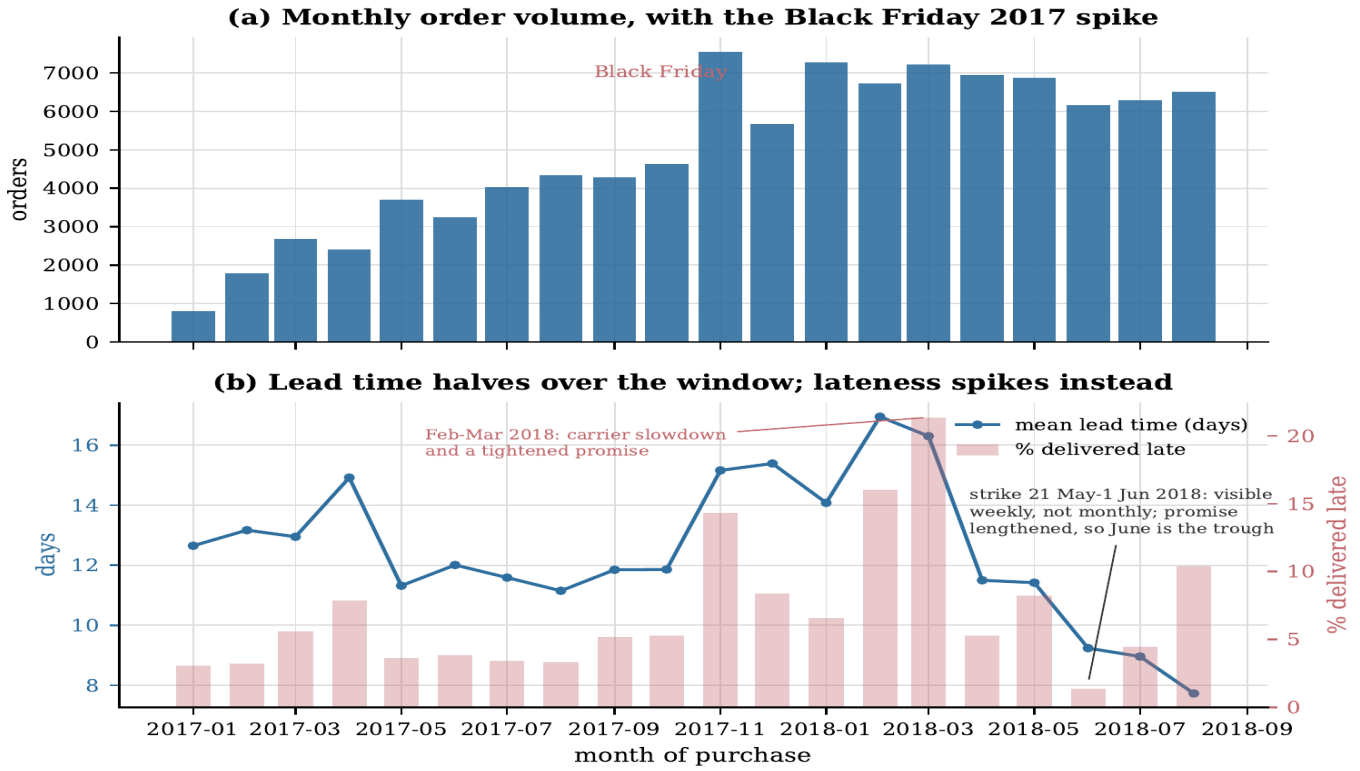
WHAT ONE FORBIDDEN COLUMN IS WORTH

0.589 → 1.000

ROC-AUC of the late-delivery model after adding the realised delivery time. That is not a result; it is the target restated — and it is why the guard is a test, not a convention.

What the data told us: drift, volatility and shocks

Three facts from the exploratory analysis that shaped every modelling decision



Monthly orders (a) and mean lead time against the late rate (b), Jan 2017 – Aug 2018

LEAD TIME DRIFTS

16.9 → 7.7 days

mean lead time, Feb → Aug 2018. Under a level shift R^2 goes negative; MAE leads, and models must be refitted on a rolling basis.

LATENESS IS VOLATILE, NOT TRENDING

1.4% – 21.4%

monthly late rate. Black Friday 14.3%; Feb–Mar 2018 a carrier-side slowdown (transit 8.7 → 13.2 d) while the promise was tightened.

THE MAY 2018 TRUCKERS' STRIKE

Only weekly

invisible in monthly data: orders in transit were 16.0% late, then the promise jumped from 19.5 to 36.8 days and June became the quietest month.

Three families as ladders — complexity has to earn its place

The same three families serve the regression and both classifiers; every rung is compared with the one below it

A · LINEAR

- rung 1 · Ridge / ℓ2 logistic
- rung 0 · LinearRegression / LogisticRegression

B · TREE-BASED

- rung 2 · HistGradientBoosting
- rung 1 · RandomForest
- rung 0 · DecisionTree

C · ENSEMBLE

- rung 1 · Stacking (A + B)
- rung 0 · Voting (A + B)

Every model is a scikit-learn Pipeline (imputation, one-hot with rare levels pooled, scaling for the linear family only). Search: RandomizedSearchCV, 30 draws, scored by five expanding-window folds. Seed 42 everywhere.

Time is respected

TimeSeriesSplit ×5 for tuning; shuffled k-fold shown only as a contrast exhibit

Select by rank, not by score

Windows differ in difficulty more than models do, so candidates are ranked within each of the 3 most recent windows

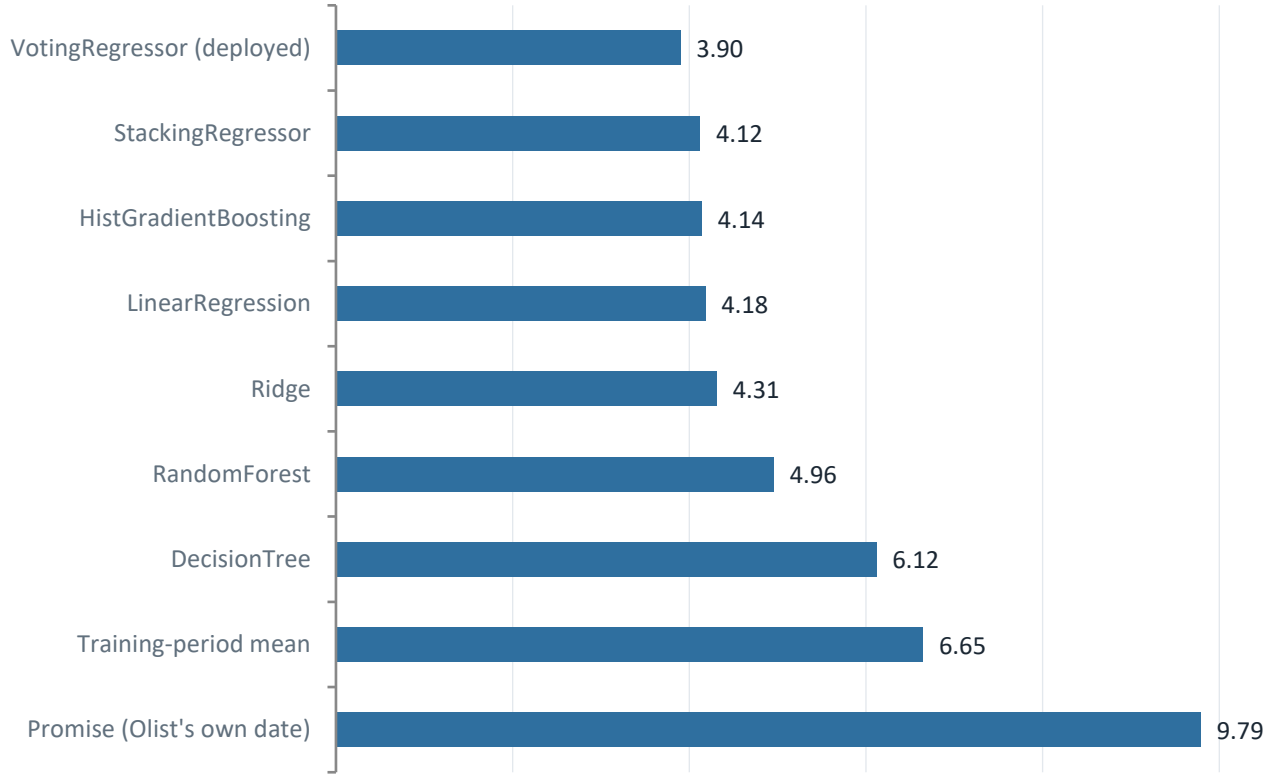
Metrics match the action

PR-AUC against its floor for choosing; Precision@K for the team's day; calibrated probabilities for the cost layer

Problem 1 — lead time, and a late flag that needs no model of its own

Held-out test window, July–August 2018; every candidate scored once

Test MAE in days, lower is better



MEAN ABSOLUTE ERROR

3.90 days

41% below a constant fitted on the past, 60% below the platform's own promised date (9.79 d)

THE LATE FLAG: DERIVED VS DIRECT

2.89× vs 1.24×

lift over the no-skill PR floor. Thresholding the regression against the promise beats every classifier trained on the binary label (ROC-AUC 0.782 vs 0.584)

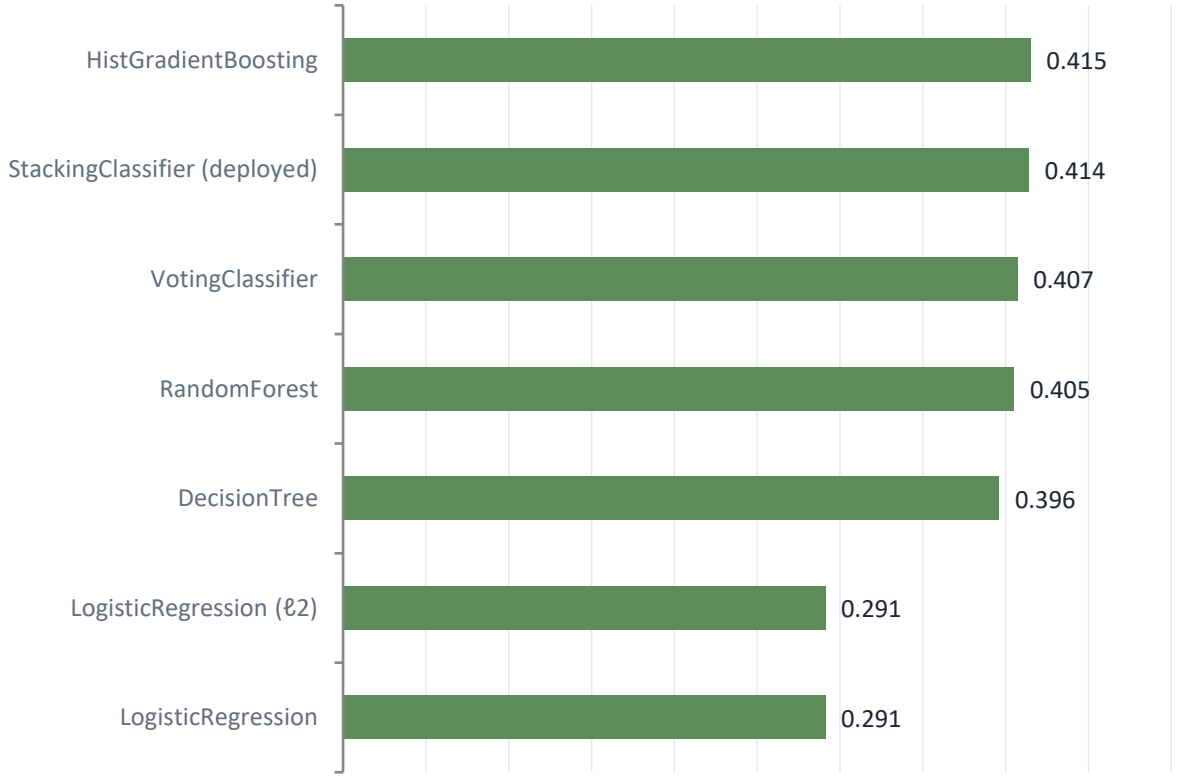
WHY THE BLEND WINS

A tree cannot predict outside the targets it was fitted on, and test-window lead times are shorter than anything common in training. The linear member extrapolates the level; averaging (1:2) lets one member cover the other's exact failure. Ridge does not beat plain least squares — regularisation bought nothing here.

Problem 2 — low-review risk at delivery

Every candidate on the held-out window, with the fold-to-fold spread the Phase 2 evaluation asked for

Test PR-AUC, higher is better · no-skill floor 0.110



ROC-AUC

0.749

balanced accuracy 0.681

PR-AUC LIFT

3.75×

0.414 vs a 0.110 floor; 0.415 on 12,617 unseen customers

FOLD-TO-FOLD VARIATION

±0.09 spread across the five folds, **< 0.001** between the best three trees. The windows differ far more than the models — so we rank within windows. Deployed: the StackingClassifier the pre-registered rule chose (mean rank 1.67 vs 2.00); the single boosted model is a statistical tie.

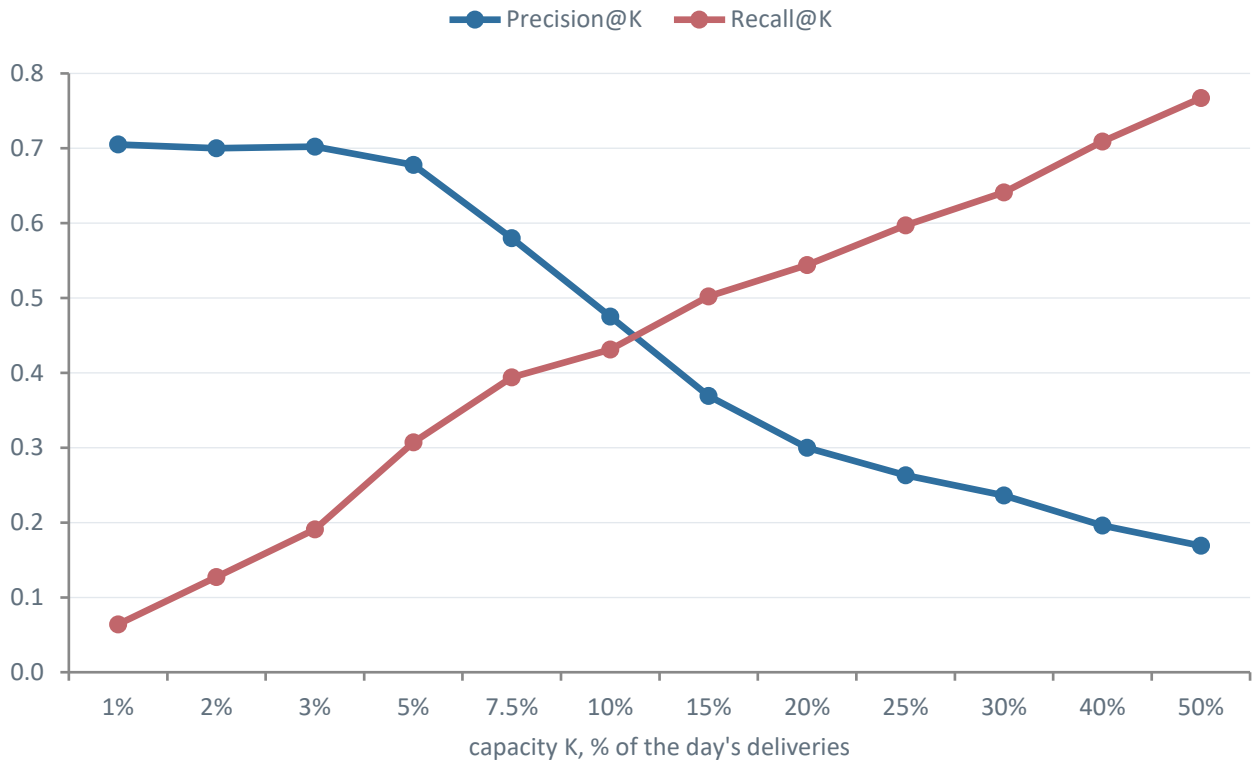
CLASS WEIGHTING: THE ABLATION PHASE 1 PROMISED

Weighting the loss: PR-AUC 0.415 → 0.411 (no gain), Brier 0.078 → 0.133, mean predicted risk 0.12 → 0.34 against an actual 0.11. It inflates every customer's risk threefold. We fit on the true prior and tune the threshold on validation instead (0.19).

Evaluate the way the team works: the top of a ranked list

A team contacts the K riskiest deliveries it can reach in a day — so Precision@K and Recall@K describe its day

Review-risk list on the test window: precision and recall of the top K% of deliveries



TOP 5% OF DELIVERIES (≈10 CONTACTS A DAY)

0.678 · 0.307

two contacts in three reach a would-be 1–2 star reviewer, covering 31% of them; cut per day, 0.572

BREAK-EVEN PRECISION (R$5 CONTACT : R$40 REVIEW)

0.125

cleared at every capacity up to half of all deliveries — capacity, not the threshold, is what binds

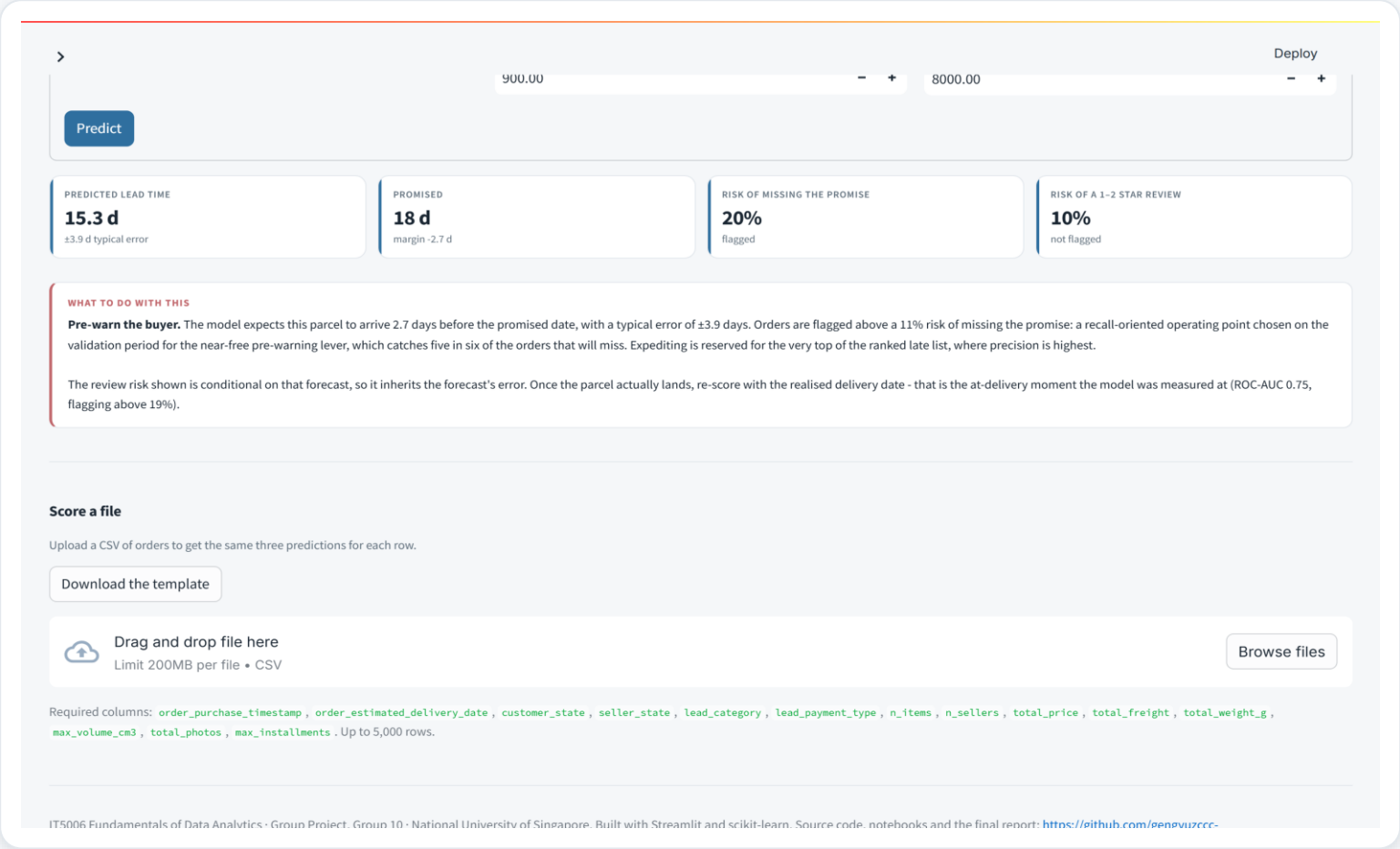
THE LATE FLAG NEVER PAYS FOR AN EXPEDITE

0.29 < 0.30

its precision peaks at 0.27–0.29 and falls with volume; a R$12 expedite needs 0.30. Informative (ROC-AUC 0.78), and still not worth paying to act on — re-quote and pre-warn instead

Deployed: team10-it5006-olist.streamlit.app

Streamlit Community Cloud · Python 3.12 · scores single orders and CSV files, returns a ranked triage list



Eight tabs

five exploratory views with filters, Predict, Replay a day, Model card

One feature pipeline

a form or CSV row runs through the same src/features.py builders the models were fitted on — nothing is reimplemented

Small and honest

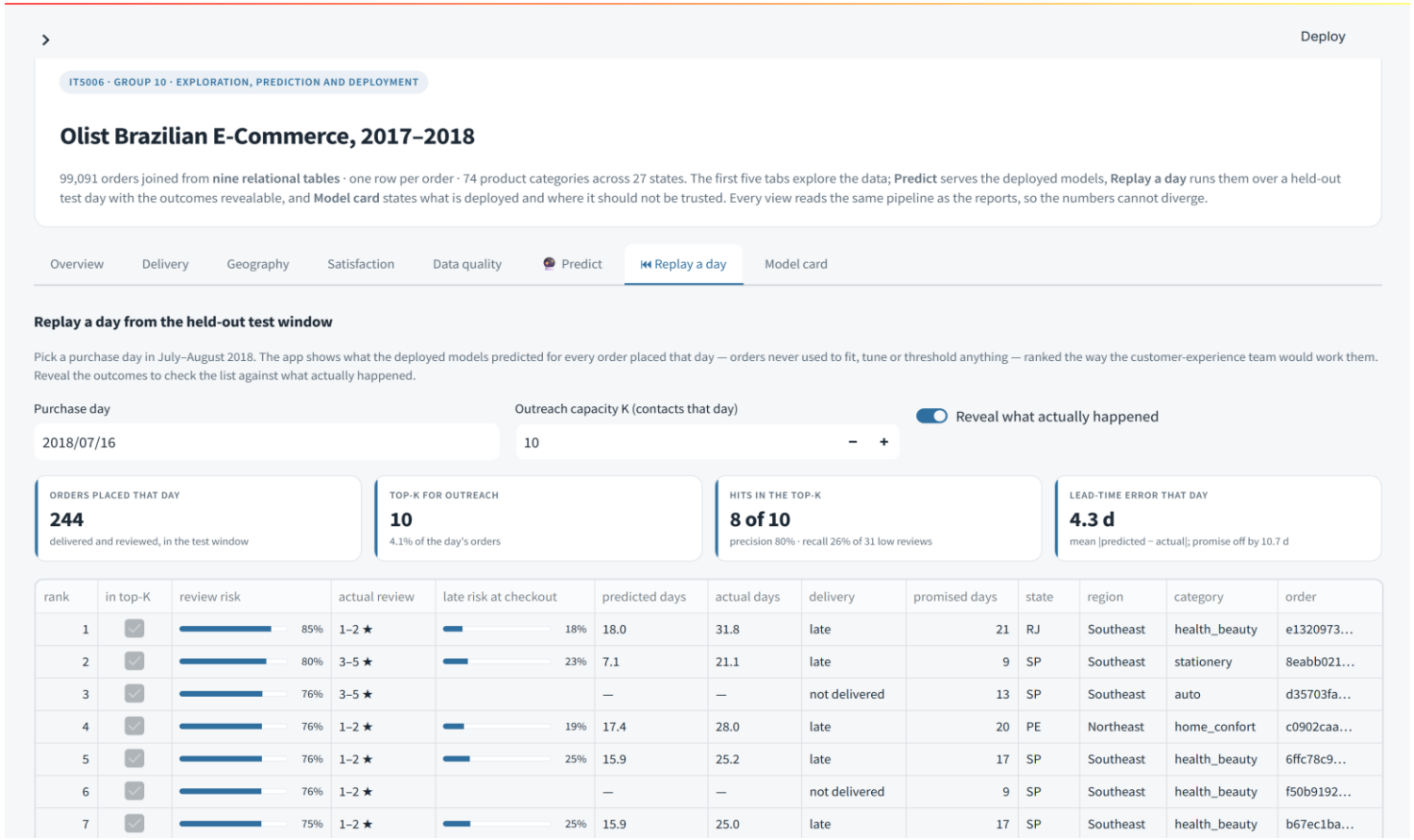
three artifacts under 1 MB, loaded lazily; scikit-learn version pinned and checked; ten edge cases handled with specific messages

Triage list

upload a day's orders, enter the team's capacity K, get the top-K marked with the precision that K achieved on the held-out window

Replay any day of the held-out window — then reveal what happened

The brief's "held-out historical records simulating new orders", made inspectable



16 JULY 2018, CAPACITY 10

8 of 10

of the ten riskiest orders placed that day, eight really left a 1–2 star review; the panel beside it shows the same capacity on all 62 test days

MODEL CARD

- What is deployed, read from the pickles: VotingRegressor (1A), the derived late flag (1B), StackingClassifier (2)
- Every searched model's selected hyperparameters
- The fold-to-fold table and chart for every candidate
- Precision@K / Recall@K; where not to trust the models
- A calculator for the outreach pilot

Recommendations: cheap levers at checkout, a ranked list at delivery

Each action with its trigger and its cost — and the experiment that would validate the business case

Who	Lever	Trigger	Cost
Operations, at checkout	Re-quote the promised date with the estimate	every order · MAE 3.90 d vs the promise's 9.79	none
	Pre-warn the buyer	late risk ≥ 0.11 · catches 5 in 6 of the orders that will miss	near zero
	Expedite freight	only where precision > 0.30 — it never is · manual review of the top 1–2% only	R$12
	Monitor sellers, routes, carrier	track-record features and the weekly late series as early warning	analyst time
Customer experience, at delivery	Contact before the review is written	the day's deliveries ranked by risk, worked top-down to capacity K	R$5
Platform	Refit monthly; monitor predicted vs realised lead time	the drift is the dominant risk to both models	—

PER 1,000 DELIVERED ORDERS

R$1,466

net benefit of the review-risk list at the cost optimum (flagging 15%); a 5% capacity still captures R$1,107



THE PILOT THAT TESTS THE ASSUMPTION

- The reais figure assumes a contact prevents the review it reaches — nobody in this data was ever contacted
- Randomise inside each day's top-K: contact half, leave half; compare the 1–2 star rate
- Control rate is already known: 57% at ten a day. Break-even: avert 1 review in 5 (21.9%)
- Six weeks at ten contacts a day detects a 1-in-4 effect (191 per arm, 80% power)
- Primary outcome: the 1–2 star rate in each arm; secondary: the realised benefit per contact, which replaces the assumed R$40

Limitations — and what we learned about method

Stated before the numbers were, and carried into the deployment plan

A moving target

Lead time fell 5.3 days between training and test; the regression over-predicts everywhere, most in the North (+4.8 d). Refit monthly; recalibrate with it.

Shocks no order can see

Black Friday, a carrier slowdown, a national strike — carrier capacity and promising policy, not order attributes. The late flag's ceiling is real.

A benefit that is assumed

The cost analysis values a contact at the review it prevents; the data contains no contacts. Pilot before scaling.

A rating is not satisfaction

Reviews are self-selected and J-shaped; the quietly dissatisfied customer never appears. Triage scores, not entitlements — never to penalise a seller.

FOUR METHOD FINDINGS THAT TRAVEL BEYOND THIS DATASET

- Cross-validation inside a drifting period reorders the candidates against the forward window, and its fold-to-fold spread is wider than the gaps between them — select by rank on recent windows, and report when even that fails (it did, once).
- Weighting the loss for imbalance leaves ranking unchanged and inflates risk threefold — move the threshold instead.
- Raw PR-AUC is not comparable across periods with different base rates; its lift over the floor is.
- The most useful business results were negative: an informative flag that does not justify an expensive action, and a policy worth testing before it is worth anything.

Thank you

Two prediction moments, one decision chain, a deployed application — and an experiment to run next.

Live application

https://team10-it5006-olist.streamlit.app

Repository, notebooks, report

https://github.com/gengyuzccc-cell/Team10_IT5006_Ecommerce_Analytics_AY2627Sem1

Questions?

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